
Numerical Partial Differential Equations

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Errata for the first printing. September 28, 2026

<https://epubs.siam.org/doi/book/10.1137/1.9781611978285>

- **page 222, exercise #7:** The Fromm slope should have a minus (not a plus):

with $a > 0$. Using the REA algorithm (see Algorithm 6.1), consider the Fromm slope

$$\delta_{k,\ell} = \frac{1}{2}(u_{k+1,\ell} - u_{k-1,\ell}).$$

- **page 289, Algorithm 7.14:** CG applied to $L^T ALy = L^T f$; **Line 7** The the final line should not have $L^T AL$.

Algorithm 7.14: CG applied to $L^T ALy = L^T f$

Input : A , linear system matrix

L , factorized preconditioner with $M = LL^T$

f , linear system right-hand side

u_0 , initial guess

Output : $\hat{y}_{k+1}$, $(k+1)$ th iterate with $u \approx L\hat{y}_{k+1}$

1 $\hat{r}_0 = L^T(f - Au_0)$, $\hat{p}_0 = \hat{r}_0$, $\hat{y}_0 = L^{-1}u_0$

2 **for** $k = 1, 2, \dots$

3 $\alpha_k = \hat{r}_k^T \hat{r}_k / \hat{p}_k^T L^T AL \hat{p}_k$

4 $\hat{y}_{k+1} = \hat{y}_k + \alpha_k \hat{p}_k$

5 $\hat{r}_{k+1} = \hat{r}_k - \alpha_k L^T AL \hat{p}_k$

6 $\beta_k = \hat{r}_{k+1}^T \hat{r}_{k+1} / \hat{r}_k^T \hat{r}_k$

7 $\hat{p}_{k+1} = \hat{r}_{k+1} + \beta_k \hat{p}_k$

- **page 401, last paragraph:** The notation here is inconsistent, introducing $\phi_{q,k}(x)$, which later becomes $\phi_{k,q}(x)$. The latter notation is more consistent:

For $v(x) \in \mathcal{V}^{h_x}$, the restriction to each element, $v_k(x)$, is a polynomial of degree at most p . Here, we consider a *nodal* representation of the polynomial, writing $v_k(x)$ in element Ω_k in terms of a nodal basis, $\{\phi_{k,q}(x)\}_{q=0}^p$, where the basis functions, $\phi_{k,q}(x)$, satisfy $\phi_{k,q}(x_{k,j}) = \delta_{q,j}$ at the $p+1$ interpolation points, $x_{k,0}, \dots, x_{k,p}$, in element Ω_k , as shown in Figure 9.5.2. The nodal basis functions $\phi_{k,q}(x)$ are given by the Lagrange polynomials.

- **page 402, line 9:** Missing $\wedge$ on ϕ_q :

the nodes $x_{k,q} \in \Omega_k$, $\phi_{k,q}(x)$, are related as $\phi_{k,q}(F_k(\hat{x})) = \hat{\phi}_q(\hat{x})$.

- page 402, two lines after (9.157): The function u^{h_x} should just be u :
equivalent, since we use the parts of u and v that are restricted to Ω_k , and are,...
- page 404, eqn. (9.165): The function z should depend on $\mathbf{u}_{k-1}$ and $\mathbf{u}_{k+1}$ in addition to $\mathbf{u}_k$:

$$\begin{aligned} M \frac{d\mathbf{u}_k}{dt} &= \frac{2}{h_x} (K \mathbf{f}_k(\mathbf{u}_k) + f^*(u_{k-1,p}, u_{k,0}) \delta_0 - f^*(u_{k,p}, u_{k+1,0}) \delta_p), \\ &=: -z(\mathbf{u}_{k-1}, \mathbf{u}_k, \mathbf{u}_{k+1}), \end{aligned}$$

- page 410, eqn. (9.187): The sign on the exponent should be positive, not negative:

$$\tilde{f}_{q,q} = \begin{cases} 1 & \text{for } q < p_{\text{cutoff}}, \\ e^{\alpha \left(\frac{q+1-p_{\text{cutoff}}}{p+1-p_{\text{cutoff}}} \right)^{2s}} & \text{for } q \geq p_{\text{cutoff}}, \end{cases}$$

- page 563, ref. [2]: The term “editors” should be removed from the citation:

R. A. ADAMS AND J. J. F. FOURNIER, *Sobolev Spaces*, vol. 140 of Pure and Applied Mathematics, Elsevier/Academic Press, second ed., 2003, <https://www.sciencedirect.com/bookseries/pure-and-applied-mathematics/vol/140>.